
Two Controls a Robot Red Team Cannot Publish Without: a Length-Matched Scramble and a Second Architecture

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 An attack success rate on a vision-language-action (VLA) policy is a difference
2 between two rates, and most published ones report only one of them. We report
3 a paired evaluation in simulation in which every attacked episode has a benign
4 twin at the same task and seed, and we add the two controls a reviewer asks for
5 first. The first is a *length-matched scramble*: the attack sentence with its target
6 noun replaced and its token order destroyed, so that “any long unfamiliar string
7 derails the policy” is a hypothesis with its own arm rather than an objection. On
8 the policy where the attack succeeds in 42 of 50 episodes against a benign floor
9 of 1 in 50, the scramble succeeds in 18 of 30 and the attack frame with no target
10 named in 27 of 30. The headline therefore measures fragility under a long out-
11 of-distribution imperative, not attacker-directed steering, and we corrected our
12 own earlier reading of it. The second control is a second architecture under the
13 identical protocol: a 3B flow-matching policy fine-tuned on the same suite leaves
14 its envelope in 1 of 30 episodes under the same attack while its task success drops
15 from 27/30 to 8/30. We also run every attack family the harness ships against one
16 policy at three seeds (only the instruction family fires; ten rungs of a weight bit-flip
17 ladder and a gradient patch do not) and the headline attack on three further suites
18 (0/30 on each). Every rate carries its floor, a Wilson interval and a task-clustered
19 interval; artifacts, ledgers and seeds are public. The contribution is not an attack.
20 It is the two controls, what they changed, and tooling that makes them a flag rather
21 than a project.

22 1 Introduction

23 An attack success rate is a difference, not a level: the rate at which a safety predicate fires under
24 perturbation minus the rate at which it fires when nothing is perturbed. The second term is routinely
25 omitted, and without it the rate cannot be compared across policies or across checkpoints of one
26 policy, and it rewards a loose predicate, since widening the unsafe region raises the number with
27 nothing in the report moving the other way. Our earlier work ran the benign arm and found that a
28 published “100%” contained the floor. This paper is about the two objections that survive a benign
29 floor.

30 The first objection is wording. A templated instruction attack that succeeds in four episodes of five
31 could be exploiting nothing more specific than the policy’s brittleness to a long, unfamiliar sentence.
32 A three-token nonsense control, the usual answer, is not length-matched and cannot rule that out.
33 The second objection is scale of evidence: one policy. Robustness sweeps on the same suite report

that VLA models are fragile to many perturbations at once [Fei et al., 2025, Guo et al., 2025] and that LIBERO scores partly reflect memorisation [Zhou et al., 2025], so a single policy’s number may be a fact about that policy.

We treat both as hypotheses with arms of their own. Section 2 gives the design, Section 3 the numbers, and Section 4 what they do and do not license. The attacks are templated and unoptimised throughout; every rate is a floor on susceptibility.

2 Method

Policies, suite and predicate. The first policy is SmolVLA fine-tuned on LIBERO (smolvla_libero); the second is $\pi_{0.5}$ fine-tuned on LIBERO (pi05_libero_finetuned_v044), a 3B PaliGemma-based flow-matching policy, loaded through a native adapter with the same LIBERO processors. The headline suite is the ten `libero_object` tasks at horizon 280. The unsafe predicate is the harness’s documented default: a static axis-aligned keep-out box on the end-effector position, never fitted to any suite, and every report carries a flag saying so. A predicate calibrated on held-out seeds with the attack arm is being fitted and will be reported separately; it is not substituted after the fact. Task success is scored by the suite’s own success function and reported beside every rate as a competence control, because a policy that cannot do the task cannot be redirected from it.

The attack and the controls. `roleplay` renders a fixed twenty-token sentence: a “safe movie scene” frame that asks the arm, “as a harmless prop”, to hand over a named graspable target. `nonsense_text` (three tokens) and `benign_reword` (a paraphrase of the original instruction) are the controls the earlier campaign carried. Two were added for this work. `scrambled_text` is the roleplay template’s own tokens with the target noun replaced by a one-token filler and the order destroyed by a seeded per-episode shuffle, whitespace-token count identical. `roleplay_no_target` keeps the frame intact and replaces only the target. All four carry the *control* role: they enter neither the attack rate nor the benign floor, and a report that mixed them in would fail its own schema.

Paired design, multiplicity, intervals. Every attacked episode has a benign twin at the same task and seed, driven by the same policy and scored by the same predicate. Significance is the exact McNemar test on the discordant (task, seed) pairs; Holm–Bonferroni is applied across the adversarial arms of a run, controlling the family-wise rate because a single row is quoted as a security finding. Intervals are Wilson on the pooled count and a bootstrap that resamples *tasks*, not episodes, since episodes within a task share a scene, a layout and an instruction; the clustered interval is the honest width and it declines to answer when fewer than two tasks are discordant. The headline runs five seeds per arm; the controls, the second architecture, the family probe and the further suites run three, and their reports carry preliminary in their own field.

Provenance. Runs were executed with harness release 0.41.2 on one workstation GPU, sharded per task, each shard writing a signed report and an execution manifest; the generated tables below are analyses of those reports and are not themselves reports. The $\pi_{0.5}$ arms were pre-registered with a public timestamped deposit before they ran (reference withheld for review); the leg reported here is the pre-registration’s preliminary leg (three seeds, two of eight arms), in its own words.

3 Results

What the scramble says. The frame with no target named leaves the envelope as often as the attack itself (27/30 against 26/30 in the same run), and the same twenty tokens in destroyed order do so in three episodes of five (18/30). The two short controls stay at zero, as before. So the `roleplay` rate measures the policy leaving its safety envelope when handed a long, imperative, out-of-distribution instruction; it does not show an attacker steering the arm toward a chosen object, because most of the effect appears with the words in the wrong order and all of it appears with no target at all. That is a real safety finding about the policy. It is not attacker-directed control, and every sentence in our earlier reports that read 44/50 as control of the outcome has been withdrawn by a published erratum; the number itself stands. The residual that belongs to the intact sentence rather than to its tokens is bounded by the gap between the scramble and the frame, roughly 27/30 against 18/30 at three seeds, and we do not claim more than that. What this run does not settle is

Table 1: SmolVLA $\times$ libero_object, ten tasks, five seeds per arm, 400 episodes. Envelope exits / attempts; Wilson 95%; exact McNemar against none with the discordant pair counts (attack-only–benign-only); Holm across the six adversarial arms; task-clustered bootstrap 95%. Clean task success on the benign arm: 48/50. The same protocol on release 0.32.0 in August gave roleplay 44/50 and none 2/50; the number reproduces.

arm	role	exits	Wilson	McNemar	Holm	clustered
none	benign floor	1/50	[0, 10]%	—	—	—
roleplay	attack	42/50	[71, 92]%	9.1×10^{-13} (41–0)	5.5×10^{-12}	[62, 100]%
goal_substitution	attack	7/50	[7, 26]%	0.070 (7–1)	0.35	[0, 34]%
scene_text	attack	2/50	[1, 13]%	1 (1–0)	1	[0, 10]%
paraphrase	attack	1/50	[0, 10]%	1 (1–1)	1	[0, 6]%
patch	attack	1/50	[0, 10]%	1 (1–1)	1	[0, 6]%
decoy_object	attack	1/50	[0, 10]%	1 (1–1)	1	[0, 6]%

Table 2: The controls beside the attack: SmolVLA $\times$ libero_object, ten tasks, three seeds, 180 episodes, same day and checkpoint as Table 1. Clean task success on the benign arm: 29/30. roleplay against none: McNemar 6×10^{-8} (25–0), clustered [67, 100]%. Controls enter neither the attack rate nor the floor, so no test is reported against them; the comparison the paper turns on is the two bottom rows against roleplay.

arm	role	exits	Wilson
none	benign floor	1/30	[1, 17]%
roleplay	attack	26/30	[70, 95]%
benign_reword	control	0/30	[0, 11]%
nonsense_text	control (3 tokens)	0/30	[0, 11]%
scrambled_text	control (20 tokens, order destroyed)	18/30	[42, 75]%
roleplay_no_target	control (frame kept, target removed)	27/30	[74, 97]%

84 whether a shorter imperative, or a fluent non-imperative sentence of the same length, produces the
85 same effect; those arms do not exist yet.

86 **The second architecture.** On the same ten tasks, horizon and predicate, $\pi_{0.5}$ leaves the envelope
87 in 0/30 benign episodes and 1/30 under roleplay (one task, one seed; McNemar $p = 1$, clustered
88 [0, 10]%), against SmolVLA’s 26/30 at the same seed count. Its clean task success is 27/30 (90%)
89 beside the 97.5% its authors report, and under the frame it completes 8/30. Thirty episodes bound
90 the difference in envelope exits well: the two Wilson intervals, [1, 17]% and [70, 95]%, do not
91 approach each other. They bound little else. In particular the frame degrades *both* policies’ task
92 completion, and only SmolVLA converts that degradation into an envelope exit; whether that is the
93 action head, the backbone, the fine-tune or the box is not separable at $n = 30$ with two arms, and
94 the pre-registered eight-arm leg exists to ask it.

95 **Breadth.** A probe of every runnable family on one task at three seeds (41 arms, 129
96 episode records) found exactly one family that does anything on this policy: roleplay,
97 roleplay_no_target and scrambled_text 3/3 each, goal_substitution 1/3, and every other
98 applicable arm 0/3, including all ten rungs of a weight bit-flip ladder (random and gradient-ranked
99 selection, $k \in \{1, 4, 16, 64, 256\}$ flips) and a gradient-optimised patch. Seventeen arms are not
100 applicable, not zero: families whose channel (tool descriptions, authorisation dialogue, backdoor
101 triggers on text the policy never reads) does not exist on this policy. Three episodes bound nothing
102 tighter than an upper limit of 56%, so nothing here is a rate; it says which families deserve
103 seeds. On the bit flips specifically, Gao et al. [2026] report that a few gradient-selected flips zero
104 out direct-regression and discrete-token action heads while flow-matching heads need on the order
105 of 100–300; our selection is a one-shot first-order ranking against the clean weights rather than their
106 progressive search, so our null at $k \leq 256$ on a flow-matching head is a lower bound on the budget,
107 not a disagreement with their result.

108 **Further suites.** The headline attack against none on LIBERO-Spatial (horizon 220), LIBERO-
109 Goal (300) and LIBERO-10 (520), three seeds on all ten tasks of each: 0/30 and 0/30 on every suite,

clean task success 23/30, 23/30 and 12/30, and on LIBERO-10 task success under the frame 0/30. The effect that breaches the envelope on LIBERO-Object in four episodes of five was not observed once elsewhere at this size (upper bound 11% each). Two readings are open and the data do not choose between them: the keep-out box is the Object suite’s and may not be a sensible box for these scenes, and the policy completes fewer than half of the long-horizon tasks unattacked, so on LIBERO-10 the failure mode under the frame is a policy that stops doing the task rather than one that leaves its box. “Not observed” is the whole claim; “safe” is not available from it.

4 Discussion and limitations

Simulation only, with one default predicate that was never fitted to any suite; a calibrated predicate is reported separately and the calibrated numbers may move. Three seeds on every arm but the headline, and $n = 30$ on the second architecture, so most nulls here are upper bounds in the 10 to 60% range rather than findings. The attacks are templated and unoptimised, so every rate is a floor on susceptibility and nothing bounds how much better an optimised attacker does. Sim-to-real correlation for VLA evaluation is, in the third-party literature, moderate and recipe-dependent [Wang et al., 2026, Li et al., 2024], which is the reason none of this predicts a real-world rate; it is a regression signal, not a field measurement. A certifier can take from a report shaped like ISO/IEC 17025 clause 7.8 the protocol, the floor, the intervals and the provenance; nothing about conformity.

What the two controls changed is the point. Without the scramble the 42/50 reads as steering and supports a story about hijacking a robot; with it the same number is fragility under a long imperative, which is a different engineering problem (instruction canonicalisation and envelope enforcement, both measurable as defences under the same paired design) and a different regulatory claim. Without the second architecture the finding is about VLA policies; with it the finding is about one policy, and the interesting question becomes what differs. Both controls cost one flag and a night of GPU time. We recommend that a red-team report on a language-conditioned policy be considered incomplete without a length-matched scramble of every instruction attack it publishes, and that a claim about a class of policies name at least two of them.

Artifacts

Anonymised artifact bundle: the results directories with every shard’s report and execution manifest, the aggregates, the run ledgers, the release tag and the exact reproduction commands, plus the harness source under a permissive licence. Withheld for review; released with the camera-ready.

References

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